UU_inference problem set — Nazifa Rahman

1. $\bar{x} = 52$, $s = 10$, $n = 100$

Test: $H_0: \mu = 50$ vs $H_1: \mu \neq 50$

$SE(\bar{x}) = \frac{10}{\sqrt{100}} = 1$ CI: $52 \pm 1.96(1) = (50.04, 53.96)$

95% confident the mean falls between 50.04 to 53.96.

We can reject the H_0.

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2. $\hat{\delta} = 3.1$, $\widehat{SE}(\hat{\delta}) = 1.3$

- $\alpha = 0.10$: CI $= 3.1 \pm 1.645(1.3) = (0.962, 5.239)$
- $\alpha = 0.05$: CI $= 3.1 \pm 1.96(1.3) = (0.552, 5.648)$
- $\alpha = 0.01$: CI $= 3.1 \pm 2.576(1.3) = (-0.249, 6.449)$

We reject null of $\delta = 0$ for $\alpha = 0.10, 0.05$, but fail to reject null for $\alpha = 0.01$, as it contains 0 inside of the interval $(-0.249, 6.449)$.

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3. $\hat{\bar{x}} = 11.5$, $s = 3$, $n = 36$

$SE(\bar{x}) = \frac{3}{\sqrt{36}} = 0.5$ CI $= 11.5 \pm 1.96(0.5)$

$= (10.52, 12.48)$

✓ yes - 12 is a plausible value because it is within the confidence interval. (95% confident the mean lies between 10.52 to 12.48. Therefore - we fail to reject the null.

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4. (A) $\bar{x}_A = 25$, $s_A = 8$, $n_A = 100$

(B) $\bar{x}_B = 22$, $s_B = 7$, $n_B = 100$

$SE(\bar{x}_A - \bar{x}_B) = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}} = 1.063$

$z = \frac{25-22}{1.063} = 2.822$

CI: $(25-22) \pm 1.96(1.063) = (0.916, 5.084)$

We are 95% confident the mean difference between neighborhoods (A) and (B) is between 0.916 and 5.084.

5. Treatment A: $\bar{X}_A = 14$, $S_A = 6$, $n_A = 81$

Treatment B: $\bar{X}_B = 12$, $S_B = 5$, $n_B = 64$

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \boxed{0.914}$$

$$z = \frac{14-12}{0.914} = \boxed{2.189}$$

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6. Factory A: $\bar{X}_A = 3.2$, $S_A = 1.5$, $n_A = 49$

Factory B: $\bar{X}_B = 2.8$, $S_B = 1.4$, $n_B = 64$

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{(1.5)^2}{49} + \frac{(1.4)^2}{64}} = 0.277$$

$$z = \frac{3.2 - 2.8}{0.277} = \boxed{1.4446}$$

$$CI = (3.2 - 2.8) \pm 1.96(0.277) = (-0.143, 0.943)$$

We are 95% confident the difference in means of factories A & B is between $(-0.143, 0.943)$ because zero is in the confidence interval → we fail to reject the null H_0, $\mu_A - \mu_B = 0$.

The difference is not statistically large relative to the noise due to zero in the 95% CI.